

Does the MOND scale track the apparent or the event horizon?

An evolving- a_0 framework, its one theory advance, and why current data cannot yet decide

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Abstract

The Milgromian acceleration $a_0 \simeq 1.2 \times 10^{-10} \text{ m s}^{-2}$ coincides with cH_0 to order unity. We take this seriously as a statement that a_0 is the surface gravity of the cosmic horizon, $a_0 = \frac{c}{2} \sqrt{G\rho} = cH/Z$ with $Z = 2\sqrt{8\pi/3}$, and ask the one question that makes it falsifiable: does a_0 track the *apparent* (Hubble) horizon, so that $a_0(z) = a_0(0)E(z)$ *rises* with redshift, or the *event* (de Sitter) horizon, so that $a_0 \propto \sqrt{\Lambda}$ is *constant*? We report an honest accounting. On the theory side we make one concrete advance: the deep-MOND *sign* requires a flat density of states, which is supplied by the (1+1)-dimensional near-horizon (s-wave) sector and not by the 3D bulk; we identify the spectrum and a candidate probe-coupling selection rule, but stop short of a derivation, which reduces to a defined calculation in the double-scaled SYK dual. On the data side, the External Field Effect—the one Λ CDM-hard MOND signature—is present in SPARC at $\sim 4.8\sigma$ and survives adversarial tests. The direct $a_0(z)$ test, however, is *inconclusive*: re-analysing the two largest real high- z kinematic samples (KMOS^{3D}, KROSS) we find the curated “rise” does not survive, the central estimates lean toward constant a_0 , but an irreducible gas+ V_{flat} systematic spans the entire framework-versus-constant gap. We conclude that the distinctive prediction is neither supported nor excluded by existing data, and that the decisive test is a dedicated deep-MOND-tail measurement at $z \simeq 1\text{--}3$ with a per-galaxy molecular+atomic gas census. We state the framework’s open core and its weaknesses explicitly, and we do not connect a_0 to the Standard Model.

1 The coincidence, taken literally

That the MOND scale satisfies $a_0 \sim cH_0$ is an old observation [1, 2]. In emergent-gravity pictures [3, 4] an infrared acceleration scale tied to the cosmic horizon is natural rather than accidental. We write the scale as the surface gravity of the free-fall horizon at the radius $R_* = c/\sqrt{G\rho}$ set by the ambient density ρ ,

$$a_0 = \frac{c}{2} \sqrt{G\rho} = \frac{c^2}{2R_*} = \frac{cH}{Z}, \quad Z = 2\sqrt{8\pi/3} \simeq 5.79, \quad (1)$$

where the second equality uses $\rho = \rho_{\text{crit}} = 3H^2/8\pi G$. With the observed a_0 this gives $H_0 \simeq 71 \text{ km s}^{-1} \text{ Mpc}^{-1}$, in the late-time camp but not competitive with the distance ladder once the unpinned factor Z (the surface-gravity “2”) and the $\sim 20\%$ stellar M/L systematic are included.

Equation (1) reproduces the standard MOND phenomenology—the baryonic Tully–Fisher relation $V^4 = GM_b a_0$, the radial-acceleration relation [5], the Freeman surface density $a_0/2\pi G$ —and we claim nothing there that MOND does not already claim. Our single new physical content is the *evolution* implied by tying a_0 to the *instantaneous* cosmic density.

The one falsifiable question. Equation (1) with $\rho = \rho_{\text{crit}}(z)$ gives $a_0(z) = a_0(0)E(z)$ with $E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$: a_0 *rises*. But the standard emergent-gravity reading ties a_0 to the cosmological constant, $a_0 \sim c\sqrt{\Lambda/3}$, which is *constant*. These are the *apparent* (Hubble) and *event* (de Sitter) horizons respectively; they agree at $z = 0$ to within $\sqrt{\Omega_\Lambda} = 0.83$, smaller than the systematic floor, and diverge by $E(3) \simeq 4.6$ by $z \simeq 3$. The entire framework reduces to: *which horizon sets the infrared scale of gravity?* Everything below is an attempt to answer it honestly.

2 The deep-MOND sign: identified, not derived

The hardest open problem in any emergent-MOND program is the *sign*: gravity must be *enhanced* at low acceleration. A temperature-modified Clausius relation gives the wrong sign (anti-MOND); the right sign comes from *freezing degrees of freedom*. On a holographic screen with $N = Ac^3/G\hbar$ bits and equipartition $Mc^2 = \frac{1}{2}N_{\text{eff}}k_B T$, freezing out a fraction $N_{\text{eff}}/N \rightarrow (a/a_0)$ below a_0 raises the temperature required to store the energy and yields $a = \sqrt{a_0 g_N}$ —enhanced, with $a_0 \sim cH$ once the freezing scale equals the de Sitter temperature $T_{\text{dS}} = \hbar H/2\pi k_B$.

What forces the freezing law. The freezing is linear, $N_{\text{eff}} \propto T$, only for a *flat* density of states $g(E) = \text{const}$: writing $g(E) \propto E^p$ gives $N_{\text{eff}}(T) \propto T^{p+1}$, and deep MOND requires $p = 0$. We computed the relevant spectra in the de Sitter static patch by the 't Hooft brick-wall method [6]. The *full* 3D bulk modes give $g(E) \propto E^2$ ($p = 2$, $N_{\text{eff}} \propto T^3$)—the area-law entropy and the correct T_{dS} , but *not* MOND. The (1+1)D *near-horizon s-wave* gives $g(E) = \text{const}$ ($p = 0$): the flat spectrum that yields the deep-MOND law. Near-horizon dimensional reduction (the membrane paradigm; the 2D conformal anomaly derivation of Hawking flux; the near-horizon CFT whose Cardy entropy is the area law) makes the relevant degrees of freedom (1+1)-dimensional.

What we do not claim. The area-law entropy lives in the 3D modes while MOND lives in the s-wave, so the two are not the same set; we offer a resolution—a spherically symmetric probe sources only the monopole ($\ell = 0$), whose sector is the flat one, so the *force* is mediated by the s-wave while the *entropy* is carried by all ℓ —but this is a selection-rule argument, not a proof, and the a_0/cH coefficient that a sharp cutoff predicts ($3/\pi^2$, i.e. $a_0 \simeq 0.30 cH$) is the right order but $\sim 1.7\times$ the observed value, so Z is *not* pinned. The honest status is: the sign is *identified* (flat (1+1)D spectrum, Gibbons–Hawking cutoff) and *motivated*, but *not derived*. The remaining steps—the equipartition projection onto $\ell = 0$ and the exact cutoff—are a defined calculation in the double-scaled SYK (DSSYK) theory that is the proposed dual of the 2D de Sitter static patch [7]; we verified one necessary condition, that the DSSYK density of states is flat near the de Sitter point.

3 The theoretical lean is conditional

Rigorous FRW horizon thermodynamics favours the apparent horizon: applying $\delta Q = T dS$ on the apparent horizon $R_A = c/H$ reproduces the Friedmann equation exactly (Cai–Kim [8]), whereas the event horizon is teleological and fails the first and generalized second laws in a general FRW

universe. Thus “ a_0 from horizon thermodynamics” selects the apparent horizon, i.e. rising a_0 . But this is a *lean, not a proof*, on two counts. First, Cai–Kim concerns the cosmic *background* (Friedmann); the link to the *local* a_0 requires a second ingredient, the Deser–Levin [9] de Sitter–Unruh floor $T(a) = (\hbar/2\pi ck_B)\sqrt{a^2 + (cH)^2}$, which saturates at T_{dS} and places the MOND knee at $a \sim cH$. Second, the constant- a_0 reading rests on a different but internally consistent premise— Λ as the fundamental infrared cutoff—which the thermodynamics does not exclude. So theory provides a coherent, two-ingredient package *for* the rising reading and does not exclude the alternative; it does not decide the question.

4 Empirical status

4.1 The External Field Effect: a robust leg

The External Field Effect (EFE) is the one signature Λ CDM does not naturally produce: because MOND is nonlinear it violates the Strong Equivalence Principle, so a uniform external field suppresses a system’s internal MOND boost [10]. In the SPARC radial-acceleration relation [11] we find the outer-disk residual turning negative (~ -0.1 dex) with growing scatter ($0.16 \rightarrow 0.26$ dex) below $g_{\text{bar}} \sim e_N a_0$ with $e_N \sim 0.03$, at $\sim 4.8\sigma$ when the significance is deflated to the number of *galaxies*. It survives adversarial tests: fixing a_0 on the high-acceleration points alone (so the downturn cannot be absorbed) leaves it at $3.7\text{--}6.0\sigma$; it appears in both random halves and survives a jackknife. The one caveat these cannot remove is the outer-disk kinematic systematic (warps, non-circular motions, beam smearing), which only a per-galaxy environment correlation settles. We regard the EFE as genuine, foundation-level evidence that gravity is Milgromian, independent of the evolution question.

4.2 The direct $a_0(z)$ test is inconclusive

A curated three-point compilation (local SPARC; Varasteanu 2025; a MUSE-DARK $z=0.9$ point) favours rising a_0 at a naive $\sim 5\sigma$, falling to $\sim 2.5\sigma$ once a redshift-growing gas/pressure systematic is marginalized, and to a coin toss if the single $z=0.9$ point is dropped. We therefore re-analysed the two largest *real* high- z kinematic samples directly.

- **KMOS^{3D}** (Übler et al. 2017 [12]; 135 galaxies, $z = 0.6\text{--}2.5$): the baryonic-Tully–Fisher $a_0 = V_c^4/(GM_b)$ is $\sim 4\times$ the local value and *rises with mass*—a high-acceleration bias—and its redshift trend is flat-to-declining, not rising. The sample is too massive and too dispersion-supported to measure a_0 cleanly.
- **KROSS** (Harrison et al. 2017 [13]; 381 rotation-dominated galaxies, $z \simeq 0.85$): with a realistic molecular-gas correction [14] the BTFR normalization gives $a_0(0.85)$ consistent with the *local* (constant) value, below the framework’s predicted 1.96×10^{-10} . Only an unphysical no-gas treatment matches the rise.

Crucially, decomposing the irreducible systematics—the CO-to-H₂ factor, the *unmeasured* atomic gas, and whether V_c reaches V_{flat} (which $a_0 \propto V^4$ amplifies)—yields $a_0(0.85) \in [0.5, 2.5] \times 10^{-10}$, *wider* than the framework (1.96) to constant (1.20) gap. The central estimate leans constant, but the systematic spans both models. The honest verdict is *inconclusive with a central preference for constant*: the curated “rise” does not survive, but the data do not exclude the framework either. No existing sample resolves this, because the gas-measured samples are massive/high-surface-brightness (wrong regime) while the kinematic samples lack per-galaxy gas.

5 Weaknesses, stated plainly

(i) *Clusters*. MOND’s residual-mass problem (a factor ~ 2 in relaxed clusters) is not relieved by the cosmic $a_0(z)$, which boosts the dynamical mass only as $\sqrt{E(z)}$ ($\sim 8\%$ at $z = 0.3$) and cannot be made environment-dependent without breaking the tight galaxy RAR. (ii) *a_0 universality*. A per-galaxy SPARC analysis rejects a strictly universal a_0 at high formal significance ($\chi^2/\text{dof} \sim 770$ statistical), reduced but not erased by the standard 0.13 dex systematic floor ($\chi^2/\text{dof} \sim 5$); the residual scatter correlates with mass at $r \sim 0.37$, suggestive of a density dependence but degenerate with fitting artifacts. (iii) *Second-order CMB*. Linear C_ℓ are a_0 -independent, but rising a_0 drives recombination into the deep-MOND regime (whereas constant a_0 leaves it Newtonian), amplifying a second-order, non-Gaussian imprint through the non-analytic $\mathcal{Y}^{3/2}$ term—a potential discriminator and a potential tension that a second-order Boltzmann calculation must settle.

6 The decisive test

The clean experiment is a direct a_0 measurement in a population of rotation-supported, deep-MOND-tail discs at $z \simeq 1\text{--}3$, analysed as a stellar- M/L - and Z -cancelling *ratio* $a_0(z)/a_0(0)$, with a per-galaxy molecular+atomic gas census and resolved rotation curves reaching V_{flat} . The three readings separate as $a_0(3)/a_0(0) = 4.6$ (apparent/rising), 1.0 (event/constant), 0.42 (scale-invariant vacuum [15], falling). An independent, kinematics-free cross-check is the weak-lensing prediction that the phantom mass at fixed baryonic mass scales as $\sqrt{E(z)}$ (+34% by $z = 1$), accessible to Euclid/Rubin/JWST stacking—immune to the pressure systematic though not to the gas census. We stress that these are the experiments that decide the question; existing data do not.

7 Conclusion

The surviving, defensible content is one equation, Eq. (1): the MOND scale as cosmic-horizon surface gravity, with the standard MOND phenomenology following and one new, falsifiable consequence—an evolving a_0 . The deep-MOND sign is identified with the flat (1+1)D near-horizon spectrum and a candidate coupling rule, but not derived; the derivation reduces to a defined DSSYK calculation. The apparent-horizon reading is theoretically favoured but only conditionally. Empirically the EFE is a robust, Λ CDM-hard detection, while the distinctive evolving- a_0 prediction is currently *undecided*: the real high- z data are gas-systematics-limited, with central values leaning constant. We have deflated the framework’s own claims—favourable and unfavourable—wherever they overreached, and we connect a_0 to gravity alone and not to the Standard Model. Whether the infrared scale of gravity tracks the apparent or the event horizon is, on the present evidence, an open and sharply testable question.

References

- [1] M. Milgrom, *ApJ* **270**, 365 (1983).
- [2] B. Famaey & S. McGaugh, *Living Rev. Rel.* **15**, 10 (2012).
- [3] T. Jacobson, *PRL* **75**, 1260 (1995).
- [4] E. Verlinde, *JHEP* **04**, 029 (2011).
- [5] S. McGaugh, F. Lelli & J. Schombert, *PRL* **117**, 201101 (2016).

- [6] G. 't Hooft, *Nucl. Phys. B* **256**, 727 (1985).
- [7] V. Narovlansky & H. Verlinde, arXiv:2310.16994 (2023).
- [8] R.-G. Cai & S. P. Kim, *JHEP* **02**, 050 (2005).
- [9] S. Deser & O. Levin, *Class. Quantum Grav.* **14**, L163 (1997).
- [10] K.-H. Chae et al., *ApJ* **904**, 51 (2020).
- [11] F. Lelli, S. McGaugh & J. Schombert, *AJ* **152**, 157 (2016).
- [12] H. Übler et al., *ApJ* **842**, 121 (2017).
- [13] C. Harrison et al., *MNRAS* **467**, 1965 (2017).
- [14] L. Tacconi et al., *ApJ* **853**, 179 (2018).
- [15] A. Maeder, *ApJ* **834**, 194 (2017).